

KHERSON, Ukraine

WMO: 339020

Lat: 46.7381N

Lon: 32.7089E

Elev: 177

StdP: 14.60

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.0	9.5	-2.4	4.9	5.7	3.5	6.6	10.2	24.6	19.0	22.6	23.2	8.0	20	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	94.4	70.2	91.1	69.2	87.9	68.1	72.9	87.0	71.3	85.1	69.9	83.3	7.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.9	106.7	77.0	67.2	100.6	75.9	65.7	95.5	75.0	36.6	87.0	35.2	85.1	34.0	83.2	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.5	18.4	16.0	-1.2	98.7	6.5	3.9	-5.9	101.5	-9.7	103.7	-13.3	105.9	-18.0	108.7		
			-1.7	76.0	6.4	1.6	-6.3	77.2	-10.0	78.2	-13.6	79.1	-18.2	80.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	28.5	31.2	39.6	51.4	62.5	70.1	74.8	73.7	63.6	51.5	41.4	32.7	(6)
(7)		DBStd	18.02	9.92	9.73	7.36	6.77	6.93	5.86	5.28	6.10	6.98	7.99	8.48	9.70	(7)
(8)		HDD50	2480	665	526	327	61	3	0	0	0	2	78	280	538	(8)
(9)		HDD65	5669	1130	945	787	409	132	20	1	5	107	423	709	1001	(9)
(10)		CDD50	3160	0	0	4	104	389	604	768	734	410	124	21	2	(10)
(11)		CDD65	874	0	0	0	2	54	173	303	273	65	4	0	0	(11)
(12)		CDH74	8972	0	0	0	28	566	1658	3162	2951	586	21	0	0	(12)
(13)		CDH80	3612	0	0	0	6	150	597	1369	1331	157	2	0	0	(13)
(14)	Wind	WSAvg	7.0	7.6	7.9	8.3	7.4	6.1	6.3	6.0	6.4	6.2	6.5	7.0	7.7	(14)
(15)	Precipitation	PrecAvg	16.9	1.3	1.2	1.2	1.1	1.7	2.0	1.7	1.3	1.5	1.3	1.3	1.4	(15)
(16)		PrecMax	26.3	2.8	3.1	2.8	2.6	5.2	3.8	5.1	4.6	4.6	5.0	2.9	3.3	(16)
(17)		PrecMin	10.9	0.2	0.2	0.1	0.2	0.0	0.6	0.0	0.0	0.2	0.3	0.0	0.1	(17)
(18)		PrecStd	4.0	0.7	0.7	0.7	0.8	1.1	0.8	1.4	1.2	1.2	1.1	0.8	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.1	54.0	63.6	77.7	88.7	93.6	98.1	99.6	88.7	76.4	64.9	55.8	(19)
(20)		MCWB	46.8	48.2	51.4	60.1	66.7	69.3	71.0	71.2	67.1	61.9	57.7	52.1	(20)	
(21)		2%	DB	46.4	49.3	58.4	71.7	84.1	89.8	93.9	95.2	84.5	71.9	60.3	51.9	(21)
(22)		MCWB	44.2	45.1	48.6	56.6	64.4	68.7	70.6	70.3	65.1	60.5	55.9	49.2	(22)	
(23)		5%	DB	43.8	46.0	54.3	67.9	80.3	86.6	91.0	91.4	80.7	67.5	56.4	48.4	(23)
(24)		MCWB	42.0	42.9	45.8	54.6	63.4	67.6	69.7	68.8	63.7	58.7	53.2	46.5	(24)	
(25)		10%	DB	40.8	43.0	50.7	64.3	76.3	83.2	87.8	87.5	76.8	63.5	53.2	45.8	(25)
(26)		MCWB	39.4	40.6	44.1	52.6	61.7	66.2	68.8	67.5	62.1	57.0	50.4	44.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.4	49.0	52.0	61.5	69.4	73.2	75.3	74.5	70.4	65.7	59.2	52.7	(27)
(28)		MDCB	49.5	52.6	61.2	75.1	84.5	86.6	89.6	91.1	80.9	72.3	62.8	55.0	(28)	
(29)		2%	WB	44.6	45.8	49.3	58.0	66.8	71.0	73.3	72.4	67.8	62.7	56.4	49.5	(29)
(30)		MDCB	46.0	48.4	56.9	69.4	79.2	83.8	87.0	87.8	78.9	68.1	59.9	51.4	(30)	
(31)		5%	WB	42.1	43.3	47.1	55.7	64.8	69.4	71.7	70.9	66.0	60.2	53.6	46.8	(31)
(32)		MDCB	43.4	45.5	52.7	65.6	76.7	81.8	85.2	85.6	74.8	65.8	56.0	48.3	(32)	
(33)		10%	WB	39.4	40.9	45.0	53.6	63.0	67.8	70.2	69.5	64.1	57.7	50.9	44.3	(33)
(34)		MDCB	40.7	42.9	49.4	62.1	74.0	79.3	83.1	83.4	73.8	63.0	53.0	45.7	(34)	
(35)	Mean Daily Temperature Range	MDBR	9.7	11.3	14.4	19.0	20.5	20.2	21.2	22.1	19.5	16.0	11.3	9.6	(35)	
(36)		5% DB	MCDBR	11.7	15.0	21.1	24.8	26.4	25.3	25.4	27.2	25.0	20.9	15.6	11.9	(36)
(37)		MCWBR	9.5	10.7	12.4	12.4	10.9	9.2	8.7	10.1	10.8	10.8	10.8	9.4	(37)	
(38)		5% WB	MCDBR	11.2	13.7	19.1	22.6	23.9	21.9	22.8	23.9	21.4	18.0	13.9	11.5	(38)
(39)		MCWBR	9.5	10.5	12.0	11.7	10.7	9.1	9.1	10.1	10.7	10.7	10.3	9.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.337	0.366	0.393	0.440	0.416	0.417	0.442	0.439	0.410	0.374	0.355	0.338	(40)	
(41)		taud	2.301	2.237	2.208	2.104	2.262	2.288	2.230	2.223	2.273	2.352	2.335	2.323	(41)	
(42)		Ebn at Noon	232	249	260	258	270	270	261	256	252	244	223	217	(42)	
(43)		Edh at Noon	25	32	39	47	42	41	43	41	36	29	24	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	331	610	1001	1528	1927	2035	2039	1833	1317	813	389	270	(44)	
(45)		RadStd	56	79	103	130	161	137	121	95	135	96	57	45	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.33	N/A	N/A	+2.20	N/A	N/A	-182	-256	+227	+148	(46)
(47)	Regional (6 neighbors)	+1.26	N/A	N/A	+1.80	N/A	N/A	-203	-275	+223	+151	(47)

Nomenclature: See separate page